



UNIT 1: ANATOMY OF A DIGITAL COMPUTER

Prachi Surlakar
Assistant Professor
Computer Engineering Department
Goa College of Engineering

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Unit - 1	Introduction to Computers: Importance of computers, characteristics of computers, classification of computers, uses of computers. Anatomy of Digital Computer: parts of computer, CPU: Control Unit and ALU. secondary storage devices, keyboards, mouse, scanners, readers, digital cameras, monitors, and printers. Operating Systems: Introduction, functions of an operating system, classification of operating systems. Introduction to Computer Problem Solving: Introduction, problem-solving aspect, top-down design.	10

PARTS OF A COMPUTER:

- **Central processing Unit(CPU) or processor:** The CPU or processor is the brain of the computer. It is the component, that actually executes instructions.
- **Memory:** Memory enables a computer to **store**, at least temporarily, data and programs.
- **Input device:** An input device usually is a keyboard or a mouse. It is channel through which data and instructions enter a computer.
- **Output device:** The typical output devices are display screens(monitors), printers, or other such devices that lets you see what computer has accomplished.
- **Mass storage device:** These devices allow a computer to permanently retain large amount of data. Example: Disk drives, tape drives

CENTRAL PROCESSING UNIT(CPU)

- The part of the **computer that executes program instructions is known as the processor or CPU.**
- In a microcomputer the CPU is on a single electronic component, the microprocessor chip within the system unit.

The CPU has two main parts

- a. Control unit**
- b. Arithmetic logic unit(ALU)**

a. Control unit

- **The control unit** tells the rest of the computer system to carry out programs instructions.
- It **directs the movement of signals between memory and ALU.**
- It also **directs these control signals between CPU and the input/output devices.**
- All computer **resources are managed** from the control unit.
- **The CPUs instructions** for carrying out commands are **built into the control unit.**
- Each instruction in the instruction set is **expressed in microcode-** a series of basic directions that tells the CPU how to execute more complex operations

The CPU has two main parts

a. Control unit

b. Arithmetic logic unit(ALU)

b. Arithmetic logic unit

- ALU performs two types of operations- arithmetic and logical.
- **Arithmetic operations**: are mathematical operations like addition, subtraction, multiplication etc
- **Logical operations**: consists of **comparisons**. Example: equal to, less than, greater than etc.
- Since **data in computer is stored in numbers**, **processing** that takes place inside the computer involves carrying out mathematical or logical operations on numbers.
- When control unit encounters an arithmetic or logic operation it passes that instruction to the ALU.
- ALU **includes** a group of **high speed memory locations** built directly into the CPU.
- These **memory locations** are called **registers** and are used to hold data that is currently being processed

MEMORY

Computers have two main types of memory: **primary memory** and **secondary memory**.

Primary Memory

- Primary memory, also **known as main memory is directly accessible by the CPU**.
- It is essential for the **computer's operation as it stores data and instructions that the CPU needs while performing tasks**.

The two primary types of primary memory are:

1. RAM (Random Access Memory):

1. **Function:** Temporarily stores data and instructions that the CPU needs during operation.
2. **Characteristics:** Volatile, meaning it loses its data when the computer is turned off.
3. **Types:** Includes DRAM (Dynamic RAM) and SRAM (Static RAM)

2. Cache Memory:

1. **Function:** A smaller, faster type of volatile memory that **stores copies of frequently accessed data from the main memory (RAM)**.
2. **Characteristics:** Located closer to the CPU, it significantly speeds up data access times.
3. **Levels:** Often includes multiple levels (L1, L2, L3) with L1 being the fastest and smallest.

ROM (Read-Only Memory): is considered part of primary memory. It is **non-volatile memory** used to store firmware and **essential system instructions that are required for the computer to boot up** and perform basic functions. Unlike RAM, which is volatile and used for temporary storage, **ROM retains its data even when the computer is turned off**.

Secondary Memory

- **Secondary memory**, also known as **auxiliary or non-volatile memory**, provides **long-term storage** for data and programs.
- Unlike primary memory, **it retains data even when the computer is turned off.**
- **Secondary storage/auxiliary storage/external storage**

There are two types of secondary storage devices based on data access:

- Sequential access** :the data stored in the media can only be read in sequence. **To get to a particular point on the media you have to go through all the preceding points.** Example: magnetic tape
- Random/Direct access**: Random access allows **data to be read or written in any order**, directly accessing any location on the storage medium without having to go through other data sequentially.
Example: **Optical disks, hard disks etc.**

Advantages of secondary storage

- **Economy**: It is more economical to store data on secondary storage devices
- **Capacity**: More capacity
- **Security**: Safe from unauthorized people or natural hazards

1. Magnetic Tapes

Magnetic tapes are a form of secondary storage medium that use a **magnetic coating on a thin strip of plastic to store data.**

They have been used for decades in various data storage applications, **particularly for backups and archival purposes.**

Characteristics

- Sequential Access:** To get to a particular point on a tape the tape must go through all preceding points.
- High Capacity:** Modern magnetic tapes **can store a significant amount of data, often exceeding the capacities of other storage devices. This makes them ideal for storing large datasets and long-term archival storage.**
- Durability:** Magnetic tapes are known for their longevity and durability. They can last for **several decades if stored properly**, making them a reliable choice for archival purposes.
- Cost-Effective:** On a per-gigabyte basis, magnetic tapes are generally more cost-effective compared to other storage media, especially for large-scale storage need.

MEMORY :Secondary Memory

2. Magnetic Disks

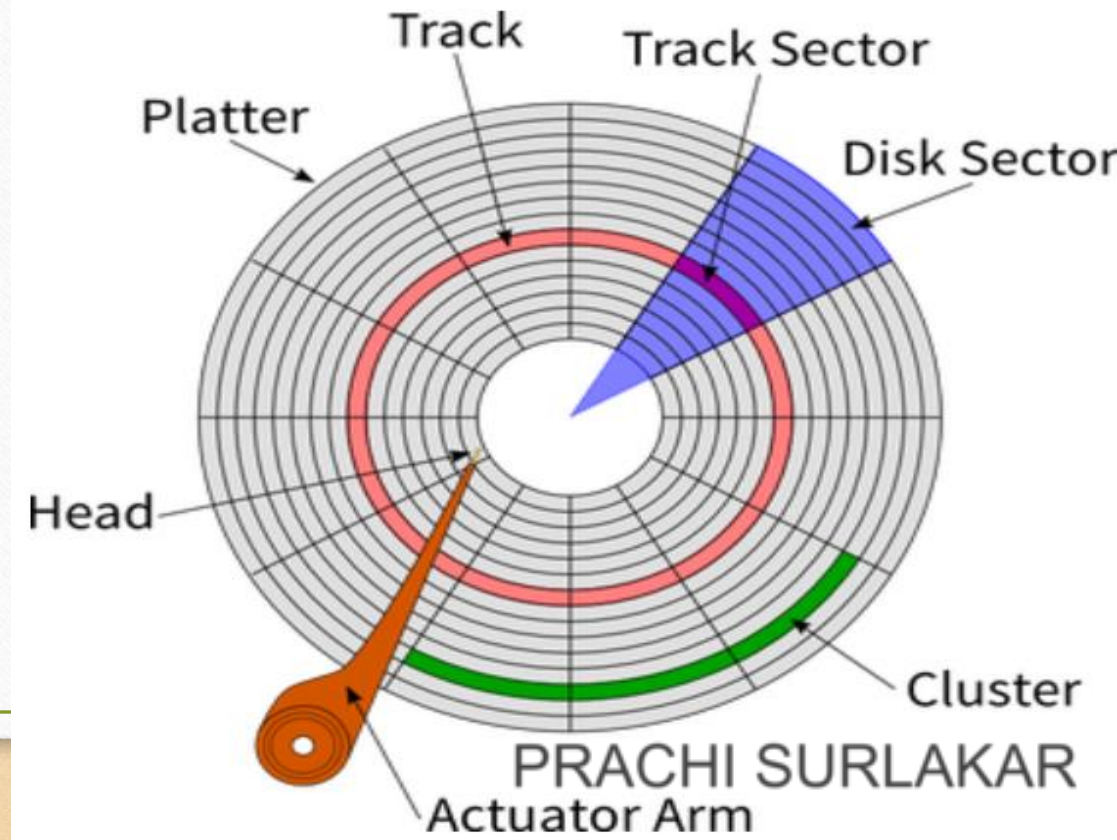
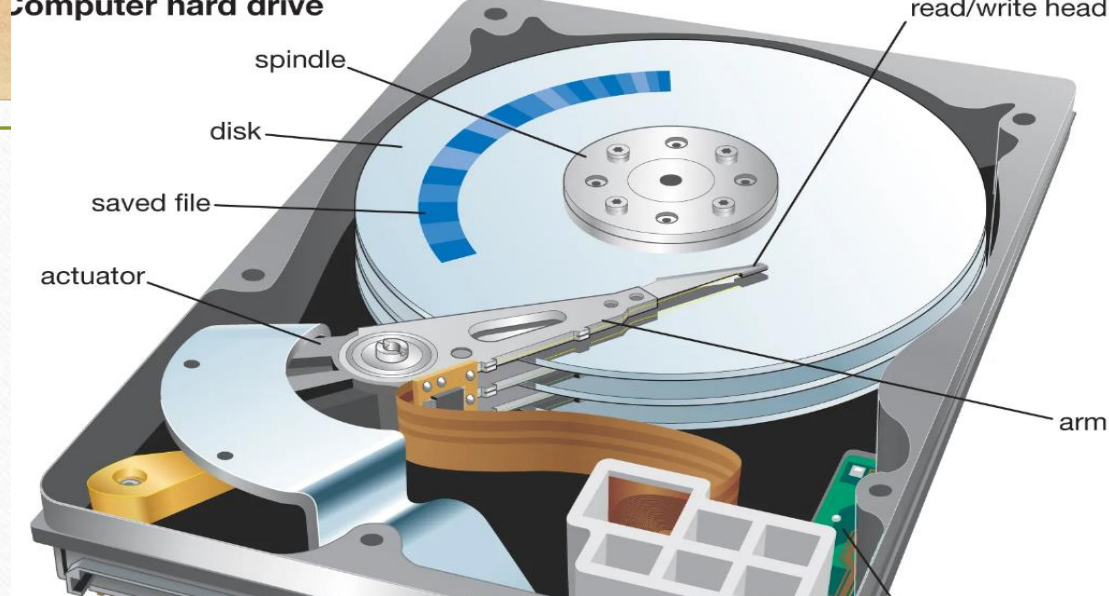
- **Direct access:** Permits direct access to data
- There are two types of magnetic disks

a) Hard disks

- b) Flexible disks(floppy disk)

a)Hard disks

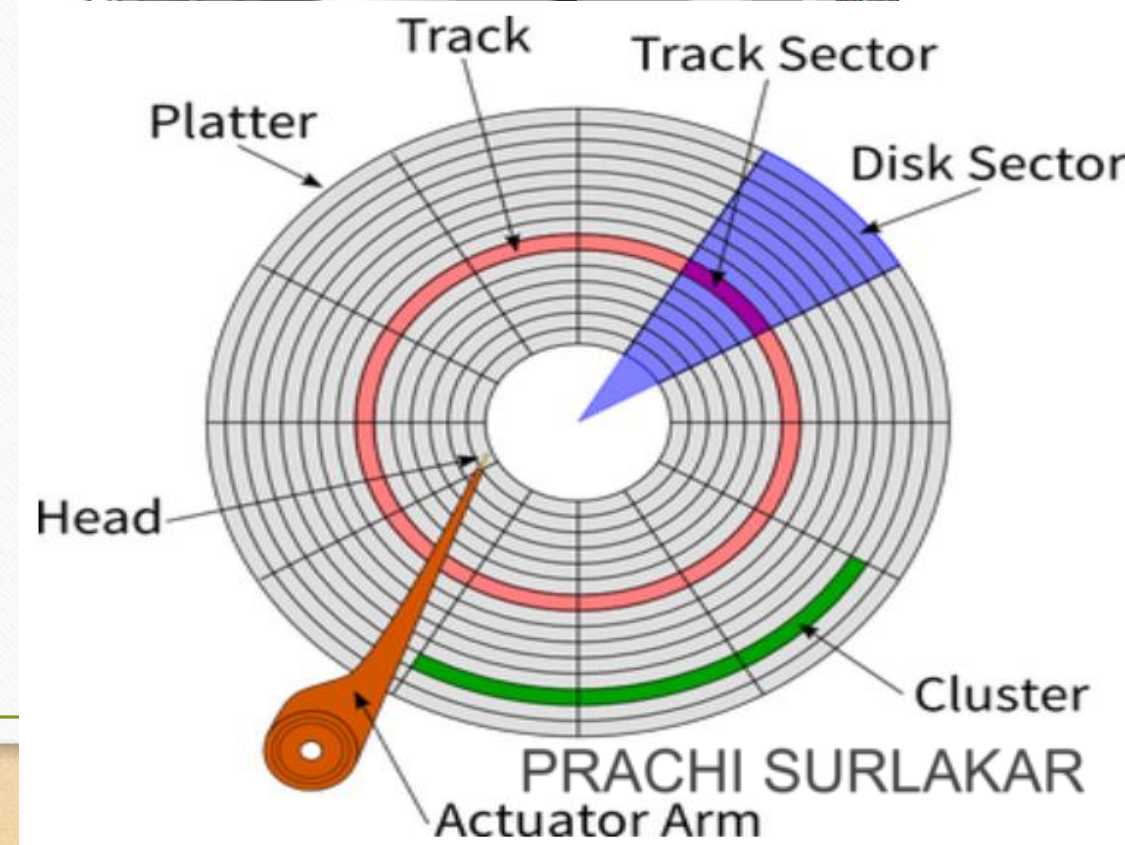
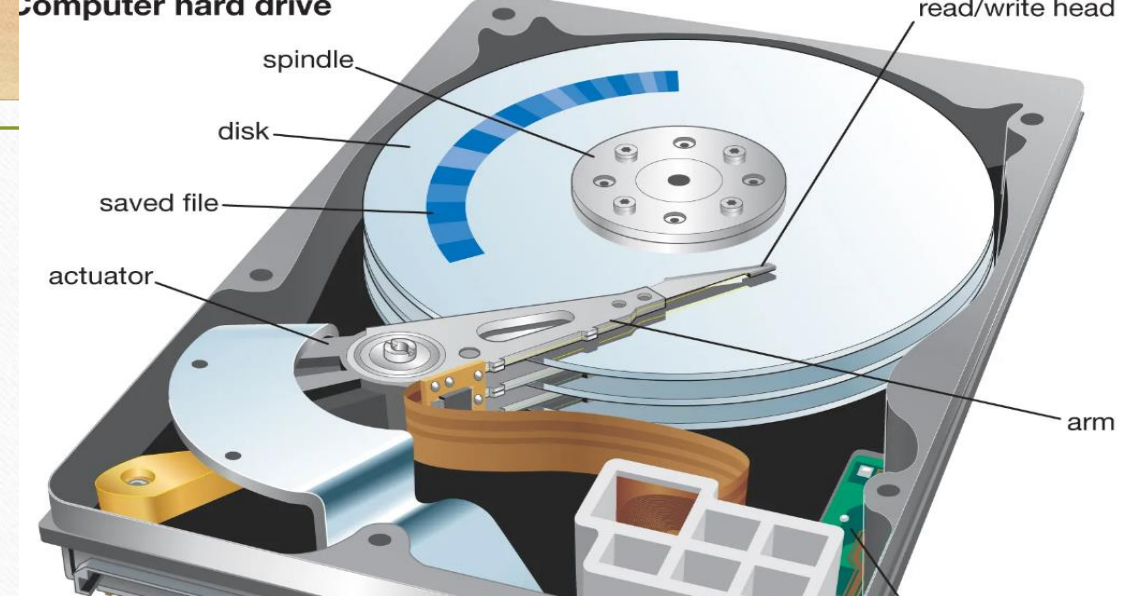
- Magnetic hard disks are thin steel platters with iron oxide coating.
- Several disks may be mounted together on a vertical shaft on which they rotate at high speeds.
- Electromagnetic **read/write heads** are mounted on access arms.
- **Data** are recorded on tracks as tiny magnetized spots forming binary digits.
- Each track can store thousands of bytes.
- **A dust particle** on disk surface can cause the head to crash into the disk called as **head crash**



MEMORY :Secondary Memory

2a)Hard disks

- Access time to locate data on hard disks is 10 to 100 milliseconds.
- Access time: is the duration taken to **complete a data transfer** -from the **time** when the computer **requests data from secondary storage** to the time when the transfer of data is complete.
- Factors of access time:
 - ❖ **Seek Time:** Time to move the read/write head to the correct track.
 - ❖ **Rotational Delay:** Time for the disk to rotate to the correct sector under the read/write head.
 - ❖ **Data transfer:** Time it takes to transfer the data to the primary storage.



2b) Floppy disks

- Floppy disk is soft magnetic disk.
- Slower to access than hard disks.
- Less storage capacity.
- Less expensive
- Portable



3. OPTICAL DISKS

- Optical disks are storage medium from which data is read and to which **data is written by lasers.**
- They have large storage capacities.
- **There are three basic optical disks.**
- **CD-ROM:** come with already data encoded into them. **Data is permanent and can be read any number of times but cannot be modified.**
- **WORM: Write once , read many.** You can write data onto WORM disk just once. After that WORM behaves like CD-ROM.
- **Erasable:** Optical disks that can be erased and loaded with new data

3. OPTICAL DISKS

CD-ROM:

recorded by vendor, once recorded they cannot be erased and filled with new data.

CD-R

Compact Disk recordable, **Write once, read many**

CD-RW

CD-Rewritable disk is a type of disk that enables you to write on it in multiple sessions

DVD

Digital Versatile Disc/ Digital video disk can hold a minimum of 4.7 GB of data

DVD-R

DVD recordable, Write once, read many

DVD-RW

DVD-Rewritable disk is a type of disk that enables you to write on it in multiple sessions

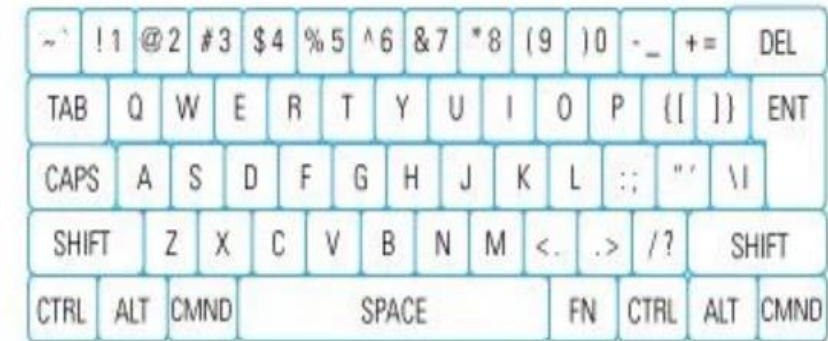
KEYBOARD

Keyboard is an input device consisting of a set of typewriter like keys that enable you to enter data into a computer.

Alphanumeric keys:

- **QWERTY:** it is called QWERTY layout because the first six keys on the top row of letters are Q,W,E,R,T,Y
- **DVORAK:** places the most commonly used letters in the middle row of keys.
- In addition to letters, alphanumeric key group consists of numbers(0-9), punctuation keys and other characters(like @, #, &, % , * etc) and few additional keys with special functions like Tab, CapsLock, Backspace, Enter and spacebar

QWERTY



DVORAK



KEYBOARD

Modifier keys:

- Modifier keys are used to **modify the input of other keys**
- You **press another key while you hold down the modifier key.**
- Modifier keys are **Shift, Ctrl and Alt.**
- Example: **Ctrl+C, Ctrl+X, Ctrl+V** are universally accepted keyboard shortcuts for copy cut and paste.

Numeric Keypad:

- Usually located at the **right side of the keyboard looks like a calculator.**
- **Contains 10 digits and four mathematical operators.**
- Also has a Num Lock key

Function Keys:

- **Function keys (F1 to F12)** are usually arranged at the top of the keyboard in a single row.
- Allow you to **input commands without typing the commands.**
- **Example: F1 is for help menu.**

KEYBOARD

Cursor Movement Keys:

- Used to navigate around the screen.
- **Arrow keys:** allows to move left, right, up and down.
- **Home and end keys:** Keys that enable you to move to the beginning or end of line.
- **Page Up and Page down keys:** enable you to move through a document screen by screen in upward and downward directions.

Special purpose keys:

- Perform special functions.
- Example: Insert, Delete, Printscreen, Start key etc.

Ergonomic keys:

- Designed with the objective of hand fatigue.
- Designed to help users hands stay positioned correctly **reducing bending and strain.**
- Keyboards may **be curved in some** way or it may be broken into **separate sections** or it **could have adjustable sections.**

KEYBOARD

Ergonomic keyboards



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KEYBOARD

- Keyboards may be wired or wireless.
- ❖ **Wireless keyboards:**
 - use **radio frequency(RF)** waves to connect to the computer).
 - They have a **receiver** which will be **connected to the computer and the keyboard sends the signals to the receiver** which is then transmitted to the computer.

MOUSE

- Mouse is a device that **controls the movement of the cursor or pointer on the display screen.**
- **Invented by Douglas Engelbart** of Stanford Research center in 1963.
- Mouse is **important for graphical user interfaces because you can point to the options and objects and click a mouse button.**
- Mouse is **useful for graphics programs that allow you to draw pictures by using mouse like a pen , pencil or a paintbrush.**

MOUSE Types of mice:

1. Mechanical Mouse:

- **Description:** Uses a **ball that rolls on a surface**, moving internal rollers to detect movement.
- **Usage:** Older technology, largely replaced by optical mice.

2. Optomechanical mouse:

- Also uses a ball to detect movement, but instead of purely mechanical sensors, it uses **optical sensors to detect the movement of the rollers**.

3. Optical Mouse:

- **Description:** Uses an **LED light and sensor** to detect movement by tracking the surface underneath.
- **Usage:** Common for general use and gaming.

4. Laser Mouse:

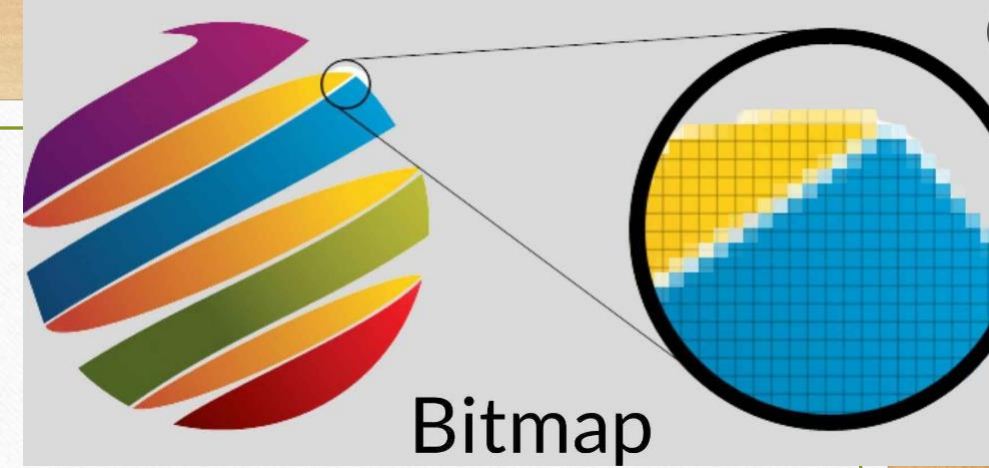
- **Description:** Similar to an optical mouse **but uses a laser** for more precise tracking.
- **Usage:** Preferred for high-precision tasks like gaming and graphic design.

5. Wireless Mouse:

- **Description:** Connects to the **computer via RF (radio frequency) or Bluetooth**
- **Usage:** Convenient for portable use and reducing desk clutter.

SCANNERS

- Scanner is an input device that can read text or illustrations printed on paper and translate the information into a form that the computer can use.
- Scanner works by **digitizing an image**-dividing it into a **grid of boxes** and representing **each box with either zero or one**, depending on whether the box is filled in.
- The **resulting matrix of bits** is called as a **bit map** that can be stored in a file, displayed on a screen and manipulated by programs.
- **Optical scanners** do not distinguish text from illustrations, they represent all images as bitmaps.
- You cannot directly edit text that has been scanned.
- To edit the text you need an **Optical Character recognition(OCR)** system to translate the image into ASCII characters



- Scanners differ from one another in the following respects:

1. **Scanning technology:**

- Most scanners use **charge coupled device(CCD) arrays** which consist of tightly packed rows of **light receptors** that can detect variation in light intensity and frequency.
- **Quality of CCD array** is the most important factor affecting the quality of the scanner.

2. **Resolution:** Denser the bitmap higher the resolution.

3. **Bit depth:** The number of bits used to represent each pixel. **Greater the bit depth more colours or grayscales can be represented.**

SCANNERS

Types of Scanners

Pen scanners:

- Scan one line at a time.
- Some advanced pen scanners can scan lines of text, store it in memory, OCR(Optical character recognition) the text and then transfer the contents to the computer.

Handheld scanners:

- Scan more area than pen scanners.
- Often called half page scanners.
- Adequate for small pictures and photos.

Flatbed scanners:

- Scan books, magazines and other documents.

Sheet fed scanners:

- Feed sheets of paper.



BARCODE READER

- Barcode readers are **photoelectric scanners** that read the barcodes, or vertical zebra striped marks printed on product containers.
- Supermarkets use a **barcode system** called **universal product code(UPC)**.
- The **barcode identifies the product to the supermarkets computer** – which has description and the latest price of the product.
- Computer **automatically tells the point of Sales(POS)terminal** what the price is.
- Barcode reader like a scanner **converts the barcode into electronic form by shining light onto the image and sensing the intensity of the lights reflection at every point.**
- Barcode reader comes in different shapes and sizes.

Handheld barcode readers:

- Reader is moved against the barcode
- Reader can also be fixed and items can be moved above it so that the barcode is read by the scanner into the computer

Slot reader

- Allows user to manually swipe a card(or other item containing a barcode) through a slot to automatically decode the barcode symbol



CARD READER

There are different types of card readers:

Business card readers:

- Business card management systems consists of business card reader that automatically reads and organizes business cards.
- The information on the card is automatically extracted and stored into their proper fields.

Smart card reader:

- Reader that Reads smart cards.
- Smart card: contains electronic memory and embedded Integrated Circuit(IC).
- Smart cards are used for variety of purposes like storing patients medical records, identity cards, passes etc.

Credit card reader:

- The **stripe** on the back of a credit card is a magnetic stripe called as magstripe.
- The **magstripe** is made up of **tiny iron based magnetic particles** in a plastic like film.
- Each particle is a **tiny bar magnet**.
- The magstripe can be written because the tiny bar magnets can be magnetized.



DIGITAL CAMERA

- A **digital camera** is a device that captures photographs and videos electronically, converting the image data into digital format.
- Unlike traditional film cameras, **digital cameras use a sensor (typically a CCD or CMOS sensor) to capture light and convert it into electronic signals.**
- These signals are then processed and stored as digital files, usually on a memory card.

Key Features:

- **Image Sensor:** The sensor is the core component that captures light. It determines the resolution of the images, measured in megapixels.
- **Lens:** Digital cameras come with either **fixed or interchangeable** lenses that focus light onto the sensor.
- **Display Screen:** Most digital cameras have an LCD screen to preview and review photos and videos.
- **Storage:** Images and videos are stored on memory cards like SD cards or microSD cards, allowing easy transfer to other devices.
- **Modes and Settings:** Digital cameras offer various shooting modes (e.g., portrait, landscape, night) and manual settings (e.g., ISO, shutter speed, aperture) to adjust how photos are captured.
- **Connectivity:** Many digital cameras feature Wi-Fi, Bluetooth, or USB connectivity for easy sharing and transferring of images to other devices.

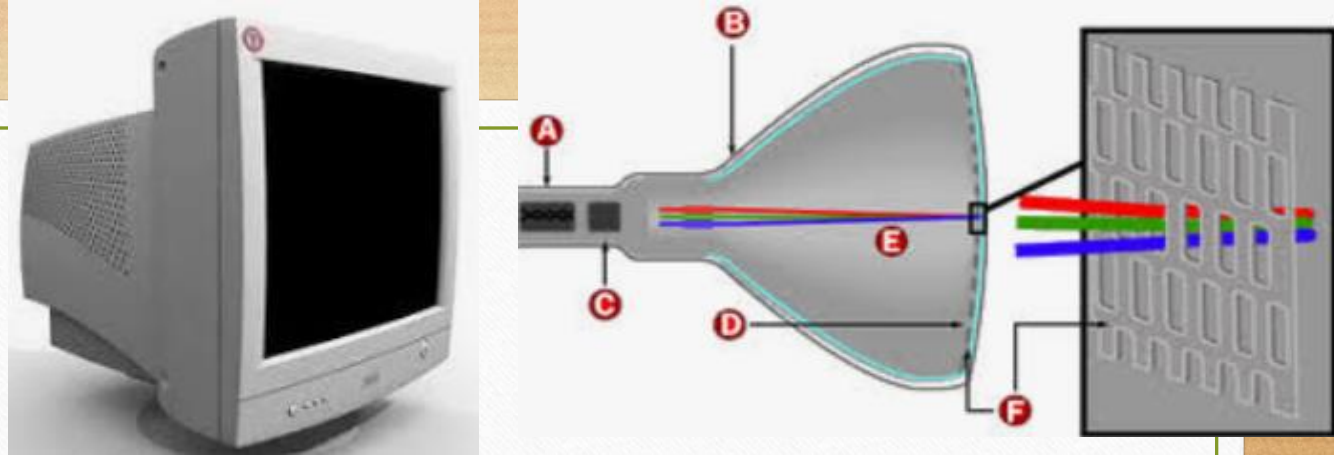
Advantages:

- **Instant Review:** Allows photographers to instantly review and delete images.
- **Editing Flexibility:** Digital images can be easily edited and enhanced using software.
- **Cost-Effective:** No need for film and developing, reducing long-term costs.
- **Storage and Sharing:** Digital files are easy to store, back up, and share online

- Most commonly used output device.
- There are two types of monitors
 - 1) **CRT monitors**
 - 2) Flat panel monitors

1) CRT monitors

- CRT monitors work by moving an electron beam back and forth across the back of the screen.
- Each time the beam makes a pass across the screen, it lights up phosphor dots on the inside of the glass tube, thereby illuminating the active portions of the screen.
- By drawing many such lines from the top to the bottom of the screen, it creates an entire screen of images.
- Near the back of the CRT monitor is an electron gun.
- The gun sends out a beam of electrons through a magnetic coil, which aims at the front of the monitor.
- In a monochrome and grayscale monitor there is only one electron gun.
- In a color monitor there are three electron guns- one for each primary color (red, green, blue).
- The color monitor displays different colors by combining various intensities of three beams.
- **Advantage:** cheap
- **Disadvantage:** Heavy, take a lot power



MONITORS

2) Flat panel monitors

2a) Liquid Crystal display(LCD monitor)

- LCD monitor creates images with a special liquid crystal that is **normally transparent but becomes opaque when charged with electricity.**

❖ **Passive matrix LCD:**

- **Relies on transistors** for each row and each column of pixel, creating a grid that defines the location of each pixel.
- The color displayed by the pixel is determined by the electricity coming from the transistors at the end of each row and top of the column.
- **Advantage:** Less expensive than active matrix
- **Disadvantage:** Narrow viewing angle, do not refresh the pixels very often resulting in poor quality.

❖ **Active matrix LCD:**

- Uses **transistor for each pixel** and each pixel is turned on and off individually.
- Allows **pixels** to be **refreshed much more rapidly.**
- LCD monitors **have wider viewing angle than passive matrix LCD.**
- Active matrix is **based on TFT (thin film transistor) displays.**
- **TFT : uses four transistors per pixel.**
- **Disadvantage:** More expensive

2) Flat panel monitors

LED Monitors:

- LED monitors are a type of LCD (Liquid Crystal Display) monitor that **uses Light-Emitting Diodes (LEDs) as the backlighting source**
- **The LED backlighting improves image quality by providing better brightness, contrast, and color accuracy.**
- LED monitors **are more energy-efficient, thinner, and lighter** compared to CCFL-backlit LCD monitors.
- They also have a longer lifespan and produce **less heat, making them more environmentally friendly.**
- LED monitors are widely used in **various applications, from desktop computers to televisions, due to their superior performance and sleek design.**

CLASSIFICATION OF MONITORS

Classification of monitors-based on colour.

1. Monochrome:

- Monochrome monitors display two colors, one for the background and one for the foreground.

2. Grayscale monitors:

- Display different shades of gray from very light gray to black.

3. Color

- Color monitors can **display anywhere from 16 to 1 million different colors.**
- Color monitors are also **called as RGB monitors because they accept three separate signals- red green and blue.**
- RGB monitor consists of a vacuum tube with three electron guns- one each for red green and blue.
- The electron guns fire electrons at the screen which contains a phosphorus coating.
- When the beam excites the phosphors they glow.

Classification of monitors-based on signals

1. Digital Monitor:

- Signal Type:** Receives a digital signal, which consists of binary data (0s and 1s).
- Connection Types:** Typically uses interfaces like HDMI, DisplayPort, DVI-D, or USB-C.
- Image Quality:** Provides a **clearer and sharper image** because there's no conversion from digital to analog, which can introduce noise or distortion.
- Usage:** Commonly **found in modern LCD, LED, and OLED displays**. Most current monitors are digital.

2. Analog Monitor:

- Signal Type:** Receives an analog signal, which is a continuous wave that represents varying voltage levels.
- Connection Types:** Typically uses interfaces like VGA (Video Graphics Array) or DVI-A.
- Image Quality:** Can be **less sharp** than digital monitors due to signal degradation, especially over long cable runs.
- Usage:** Analog monitors were more common in older CRT (Cathode Ray Tube) monitors, though some early LCD monitors also supported analog input.

CHARACTERISTICS OF MONITORS

1. **Size:** Screen sizes are measured in **diagonal inches**, the distance from one corner to the opposite corner diagonally.
2. **Resolution:** Resolution of a monitor indicates **how densely the pixels are packed**. The number of bits used to represent each pixel determines how many colours or shades of gray can be displayed.
3. **Refresh rate:** Refresh rate determines how many times per second the screen is to be refreshed(redrawn). Refresh rate is measured in Hertz(Hz)
4. **Bandwidth:** Bandwidth is the **amount of data that can be transmitted in a fixed amount of time**. For **digital** devices, the bandwidth is expressed in bits or bytes per second(**bps**).For **analog** devices, bandwidth is expressed in cycles per second or Hertz(Hz).
5. **Interlaced or non interlaced:** Interlacing is a display technique that enables a monitor to provide more resolution inexpensively.
6. **Dot pinch:** Dot pinch is a measurement that indicates the **vertical distance between each pixel on a display screen**. Lower the number crisper the image.
7. **Convergence:** refers to **how sharply an individual color pixel on a monitor appears**. Each pixel is composed of three dots- red green and blue. Three dots should converge at the same point.If the dots are badly mis converged the pixel will appear blurry.

MONITORS: VIDEO STANDARDS

Standard	Resolution	Simultaneous Colors
VGA	640 x 480	16
	320 x 200	256
SVGA	800 x 600	16
	1024 x 768	256
	1280 x 1024	256
	1600 x 1200	256
8514/A	1024 x 768	256
XGA	640 x 480	65536
	1024 x 768	256
TI 34010	1024 x 768	256

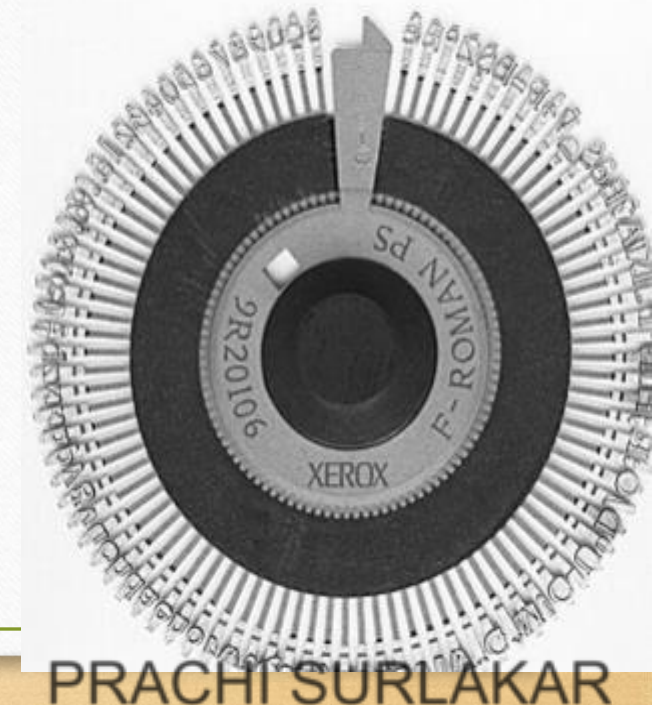
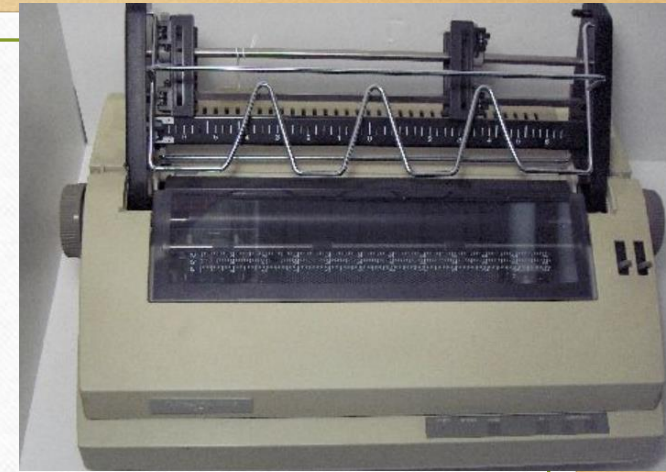
PRINTER

CHARACTERISTICS:

1. **Quality of type:** Output produced by printers is said to be either letter quality, near letter quality or draft quality.
2. **Speed:** Measured in characters per second(**cps**) or pages per minute(**ppm**).Example: Laser printers range from about 4 to 20 text pages per minute.
3. **Impact or non impact:** Impact printers include all printers that work by striking an ink ribbon. Daisy wheel,dot matrix are impact printers. Non impact printers include laser printers and ink-jet printers. Impact printers are noisy.
4. **Graphics:** Some printers (daisy wheel) can print only text. Other printer can print text and graphics.
5. **Fonts:** Some printers are limited to a few fonts. But most printers can print different fonts.

DAISY WHEEL PRINTER

- Daisy wheel printer **produces letter quality type.**
- Daisy wheel is a **disk** made of plastic or metal on which **characters stand out in relief** along the outer edge.
- To print a character, the **printer rotates the disk** until the **desired letter is facing the paper.**
- Then a **hammer strikes the disk** forcing the character to hit **an ink ribbon**, leaving an impression of the character on the paper.
- **Disadvantage:** noisy and slow
- These printers have become almost obsolete.



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DOT MATRIX PRINTER

- Dot matrix printers **create characters by striking pins against an ink ribbon.**
- Each pin **makes a dot** and combinations of dots form characters.
- **Advantage: cheap**
- **Disadvantage: noisy**
- **Characteristics:**
 - ❖ **Speed:** given in characters per second(cps), the speed can vary from 50 to over 500 cps
 - ❖ **Print Quality:** Determined by the number of pins(the mechanisms that print the dots), it can vary from 9 to 24



Ink-jet printer

- Ink-jet printers work by **spraying ionized ink at a sheet of paper.**
- **Magnetized plates in the ink's path direct the ink onto the paper in the desired shapes.**
- **Capable of producing high quality print.**
- **Advantage:** Price is lower than laser printers.
- **Disadvantage:** slower than laser printers



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Laser printer

- ❖ Laser printers utilizes a **laser beam to produce an image on a drum.**
- ❖ The **light** of the laser alters the electrical charge on the drum wherever it hits.
- ❖ The **drum is then rolled through a reservoir of toner**, which is picked up by the charged portions of the drum.
- ❖ Finally **toner is transferred to the paper** through a combination of heat and pressure.
- ❖ Because an entire page is transmitted to a drum before the toner is applied, laser printers are sometimes called page printers.
- ❖ **Monochrome laser printer:** uses single toner
- ❖ **Color laser printers:** use four toners to print full color.
- ❖ Non impact printers
- ❖ Quieter than dot matrix or daisy wheel printers.
- ❖ Good speed: 4 to 20 pages per minute



- **LCD/LED printer:** Similar to a laser printer but uses liquid crystals or light emitting diodes to produce image on the drum.
- **Thermal printers:** printers that produce images by pushing electrically heated pins against special heat sensitive paper.

THANK YOU😊

by PRACHI SURLAKAR

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